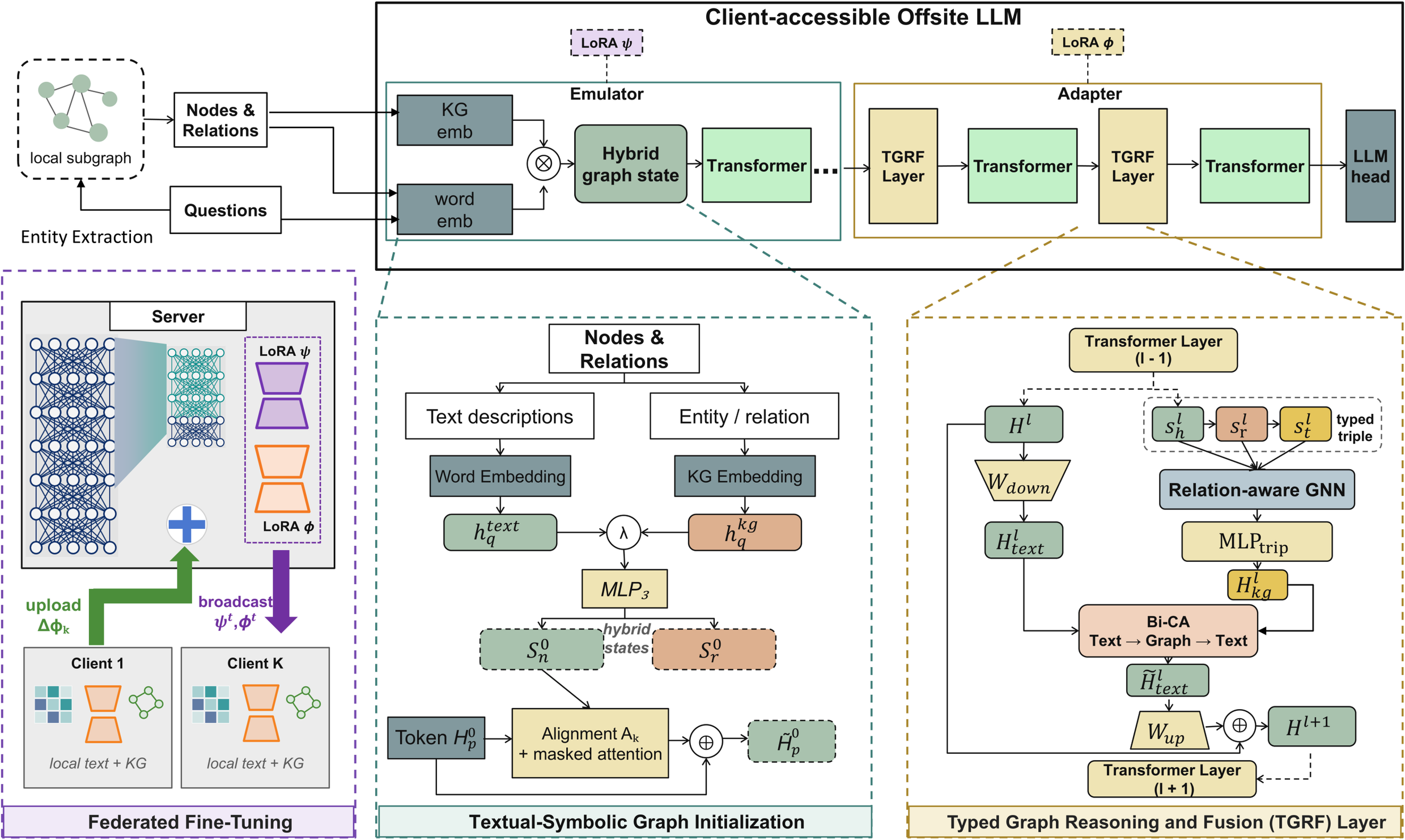

Client-accessible Offsite LLM
LoRA ψ
LoRA ϕ
Emulator
Adapter
local subgraph
Nodes & Relations
Questions
Entity Extraction
KG emb
word emb
Hybrid graph state
Transformer
TGRF Layer
Transformer
TGRF Layer
Transformer
LLM head
Server
LoRA ψ
LoRA ϕ
upload Δϕk
broadcast ψt,ϕt
Client 1
Client K
local text + KG
local text + KG
Federated Fine-Tuning
Nodes & Relations
Text descriptions
Entity / relation
Word Embedding
KG Embedding
h_q^{text}
λ
h_q^{kg}
MLP_3
hybrid states
S_n^0
S_r^0
Token H_p^0
Alignment A_k + masked attention
$\tilde{H}_p^0$
Textual-Symbolic Graph Initialization
Transformer Layer (l - 1)
H^l
W_{down}
H_{text}^l
s_h^l
s_r^l
s_t^l
typed triple
Relation-aware GNN
MLP_{trip}
H_{kg}^l
Bi-CA Text → Graph → Text
$\tilde{H}_{text}^l$
W_{up}
H^{l+1}
Transformer Layer (l + 1)
Typed Graph Reasoning and Fusion (TGRF) Layer